

rechargeable lithium-ion batteries that can last a maximum of 8 hours, and a non-rechargeable backup battery that is good for an additional 12 hours. Undoubtedly, with the kind of rapid advances in fuel cell technology, batteries are going to get lighter and last longer. Finally, to ensure that none of these hi-tech gizmos land up in the wrong hands, soldiers or the control centre can effectively destroy everything at the press of a button.

Though it may be a tad disappointing, the networked soldier of the future will still go to war with his modified M4 by his side—and not a light sabre or a phaser set on stun. The M4 is a variant of the US Army standard-issue M-16A2 rifle. The main improvement in the design is the addition of the Thermal Weapons Sight system, a daylight camera, a laser

that has warm or cold air blown in through an intricate network of capillaries. The uniform will also come equipped with multiple physiological sensors that let the soldier, as well as medics and the control centre monitor vital signs—medics will know when a soldier is hurt, and further advancements may see them diagnosing his condition, while still on the way to his location. It has also been built to protect the soldier in chemically and biologically contaminated areas. Some radi-



The Stryker—Pouncing up for...

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“ Machines don't fight wars. People do, and they use

The late Colonel John R Boyd

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rangefinder and digital compass. The optics within the M4 will also be hooked up to the helmet mounted displays—super soldiers can sight and fire at targets anywhere, even around corners, and avoid getting their heads blown away.

The clothing that will be used by the soldiers is also going to change radically—in the Objective Force Warrior program, the uniform is a multifunctional device that works from the inside out. The suit comes with microclimate conditioning,

cal new advances in camouflage have resulted in technology that prevents detection from thermal imaging, locking in the body heat. With all this technology, the face of the infantryman in battle might transform from a blood, guts and cordite stained warrior into a super-geek.

Infantry...nah, I drive Armoured

One of the US Army's most eagerly awaited armoured combat weapons system is the Stryker or the 'smart battlefield taxi'.

armoured troop carrier that can carry up to nine infantry soldiers and has a crew of two. This selectively eight-wheel-drive, and full time four-wheel-drive, armoured combat vehicle offers unbelievable tactical mobility and traction over all sorts of surfaces. It weighs 19 tons, can speed up to a maximum of 62 Mph (about 100 Kmph). The Stryker comes with three unique touch-screen flat-panel displays that give its commander all the information that he needs—there is an electronic visor that shows the troops in the rear an enhanced view of what the driver sees. Think of it as an electronic control room that monitors squad-level intelligence as well as equipment. An onboard micro-Ethernet hub from Cisco, networks all this data. The Stryker gives the troop commander the rapidly updated battleground information that he needs to wage war

It's also well hooked up for TV—the Stryker's fire control is linked to powerful cameras that include day and night vision capabilities. Through it, the battlefield commander can sight and aim at targets, and give the command to fire any weapon, including 105 mm calibre heavy guns, a MK19 40 mm grenade launcher, and even guided missiles that have been mounted on the roof of the vehicle. This is just the sort of heavily armoured, rapidly deployable troop carrier that the US Army expects to use for highly localised conflicts.



JSF-35—Looking beyond to the other side and co-ordinating with ground-based units

